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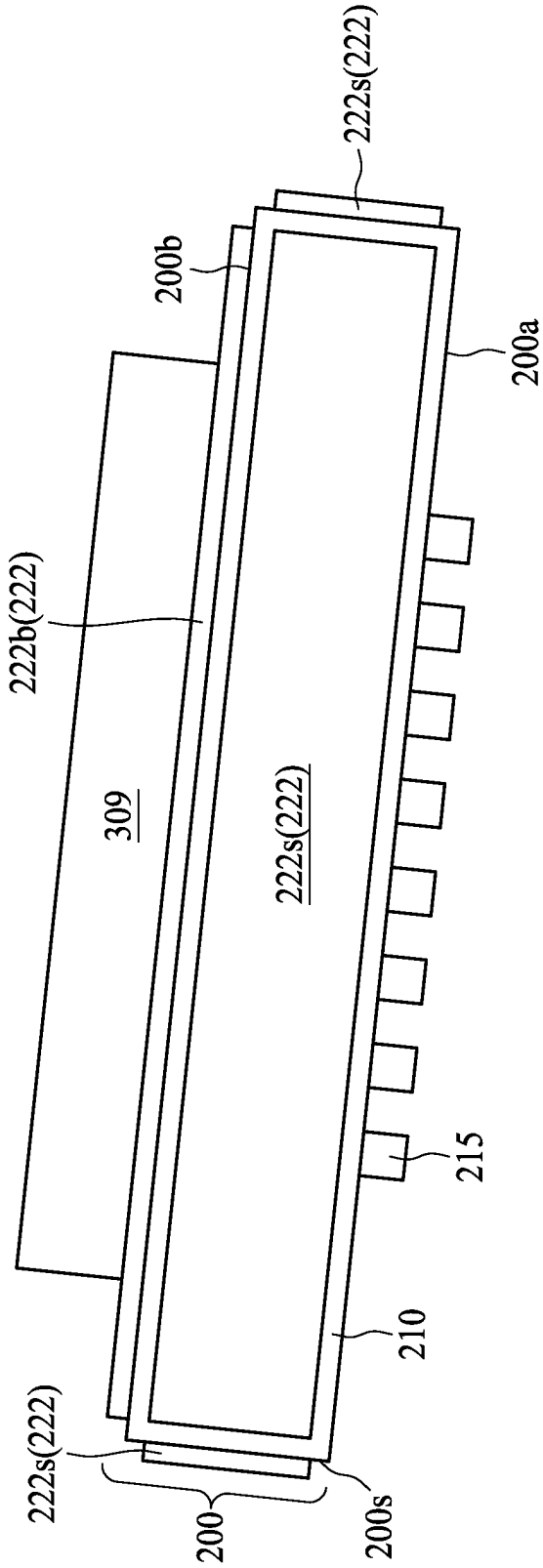


FIG. 1

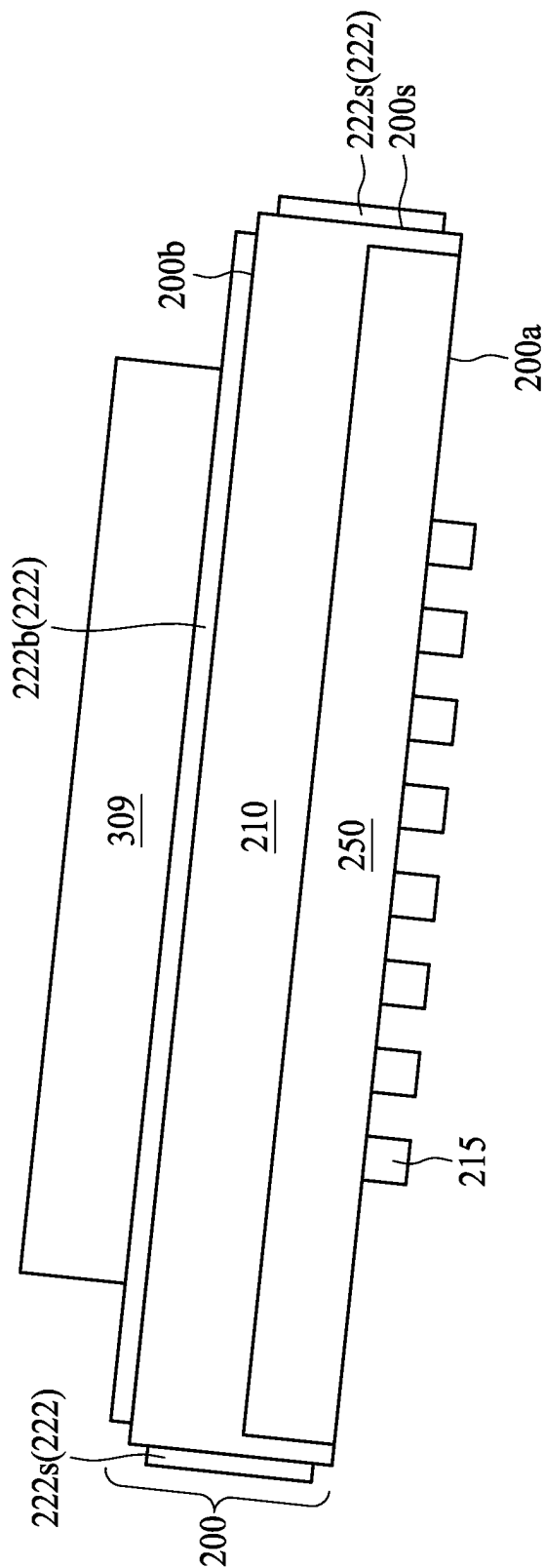


FIG. 2

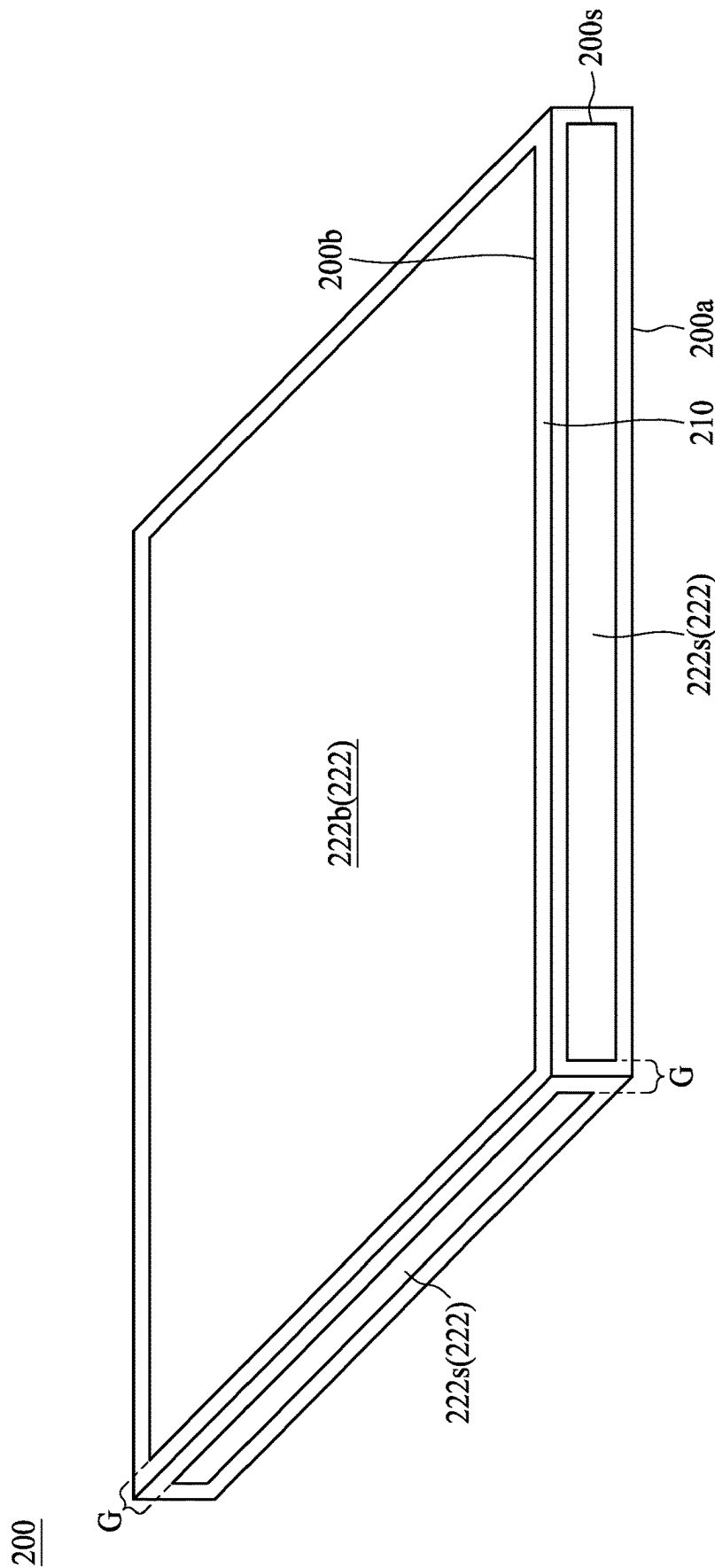


FIG. 3

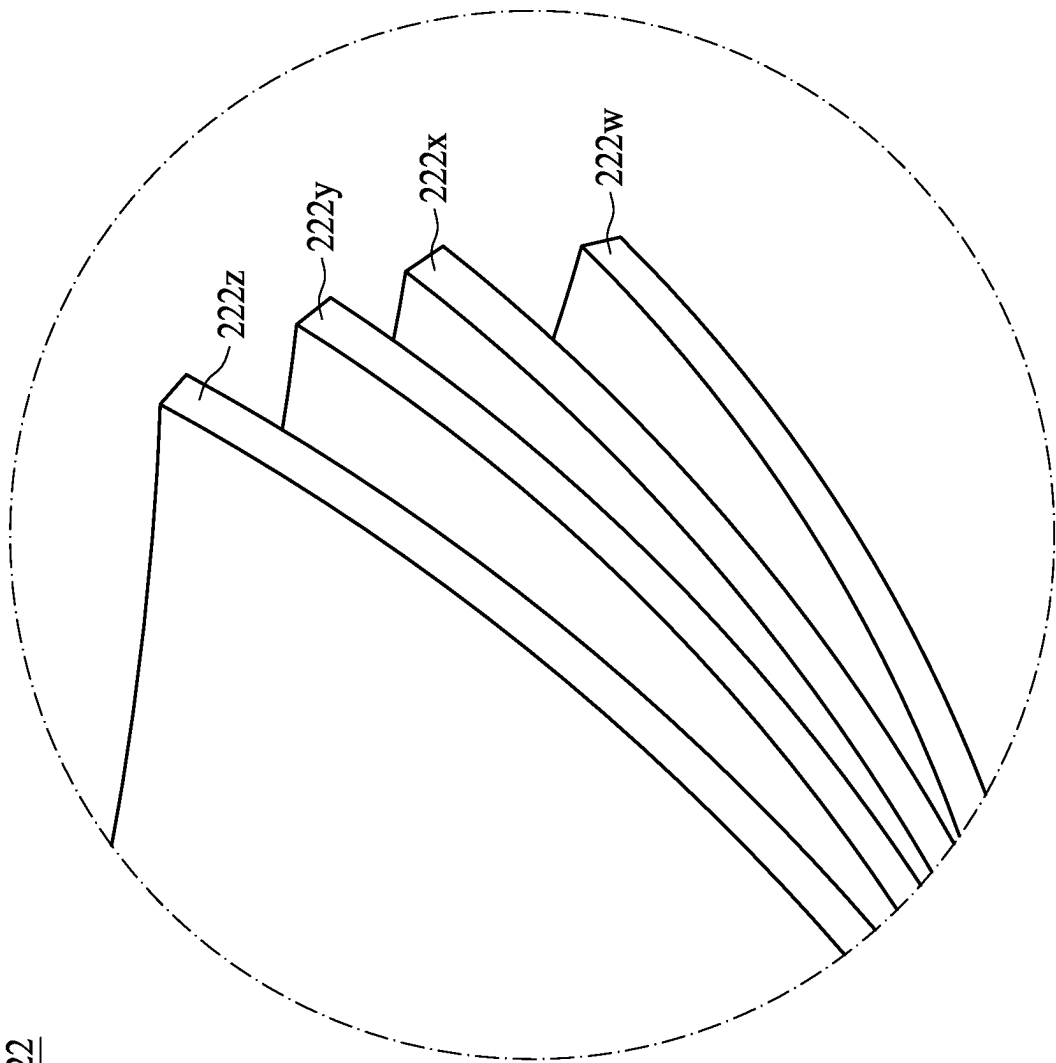


FIG. 4

222

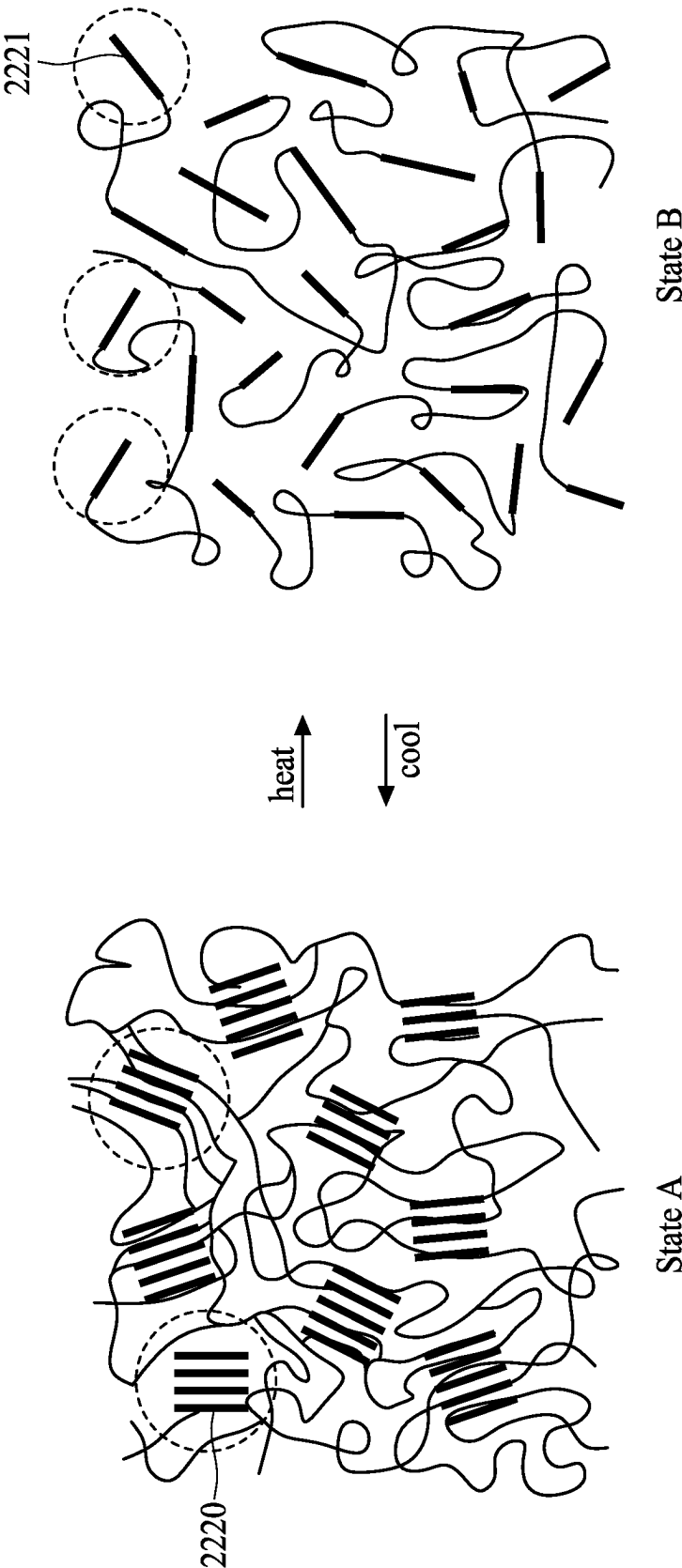


FIG. 5

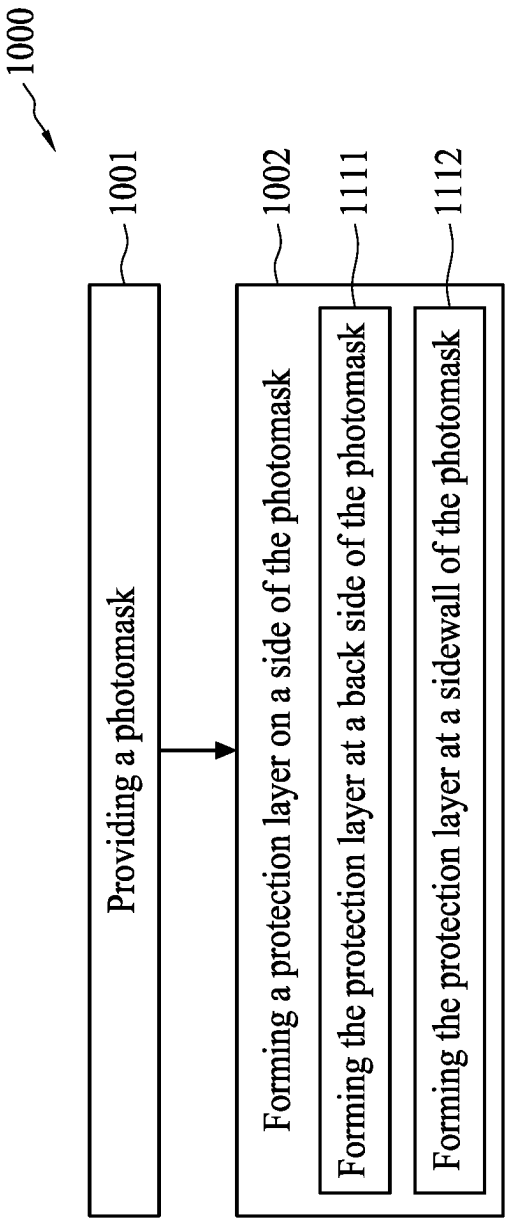
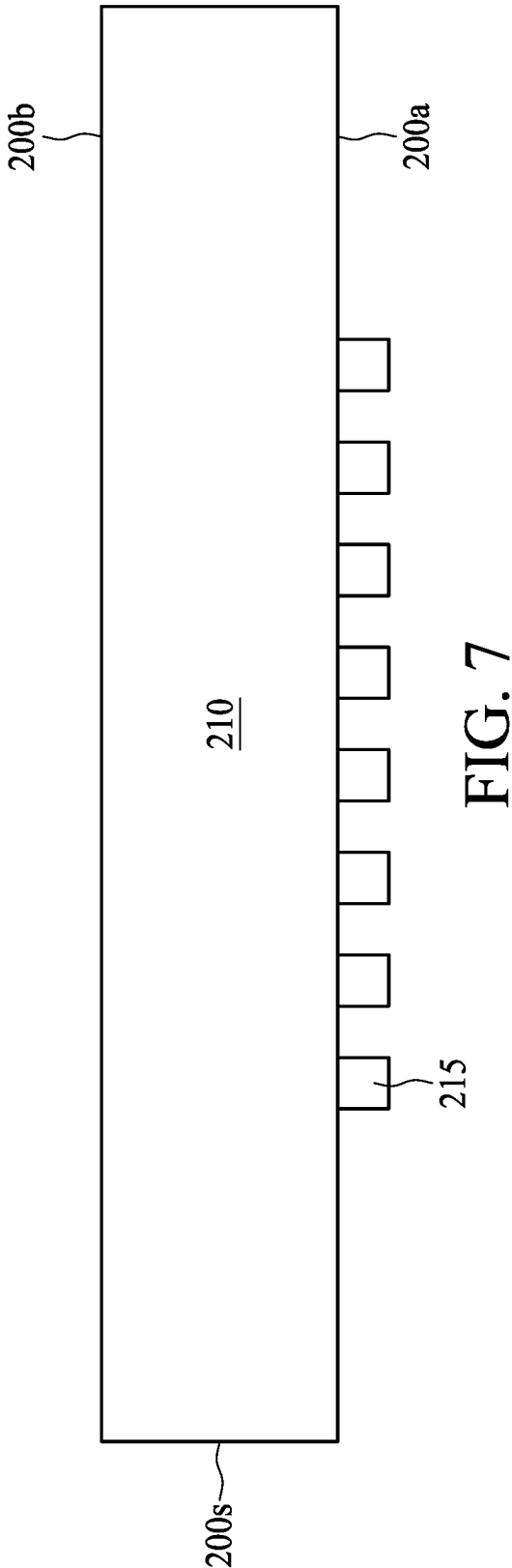
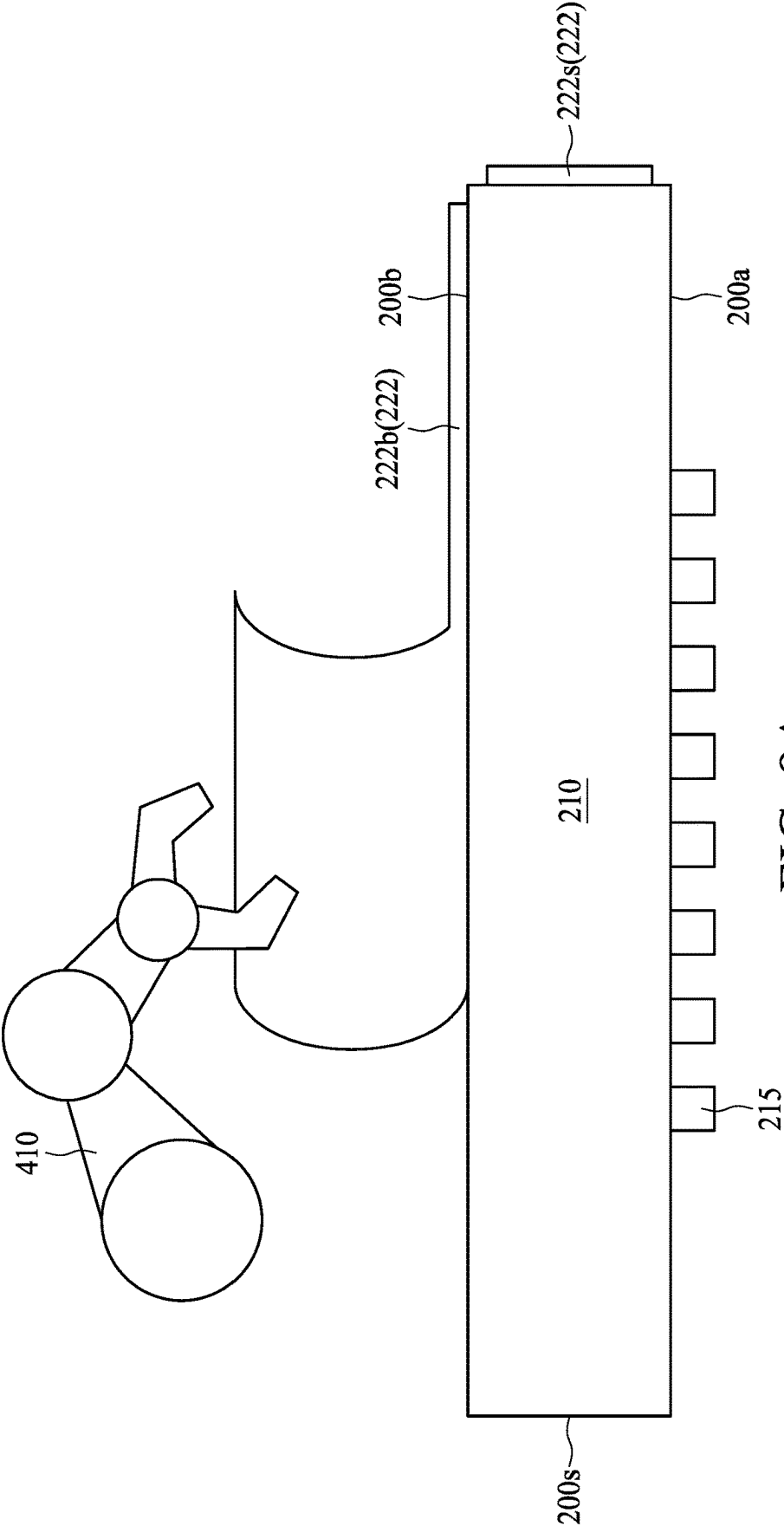


FIG. 6





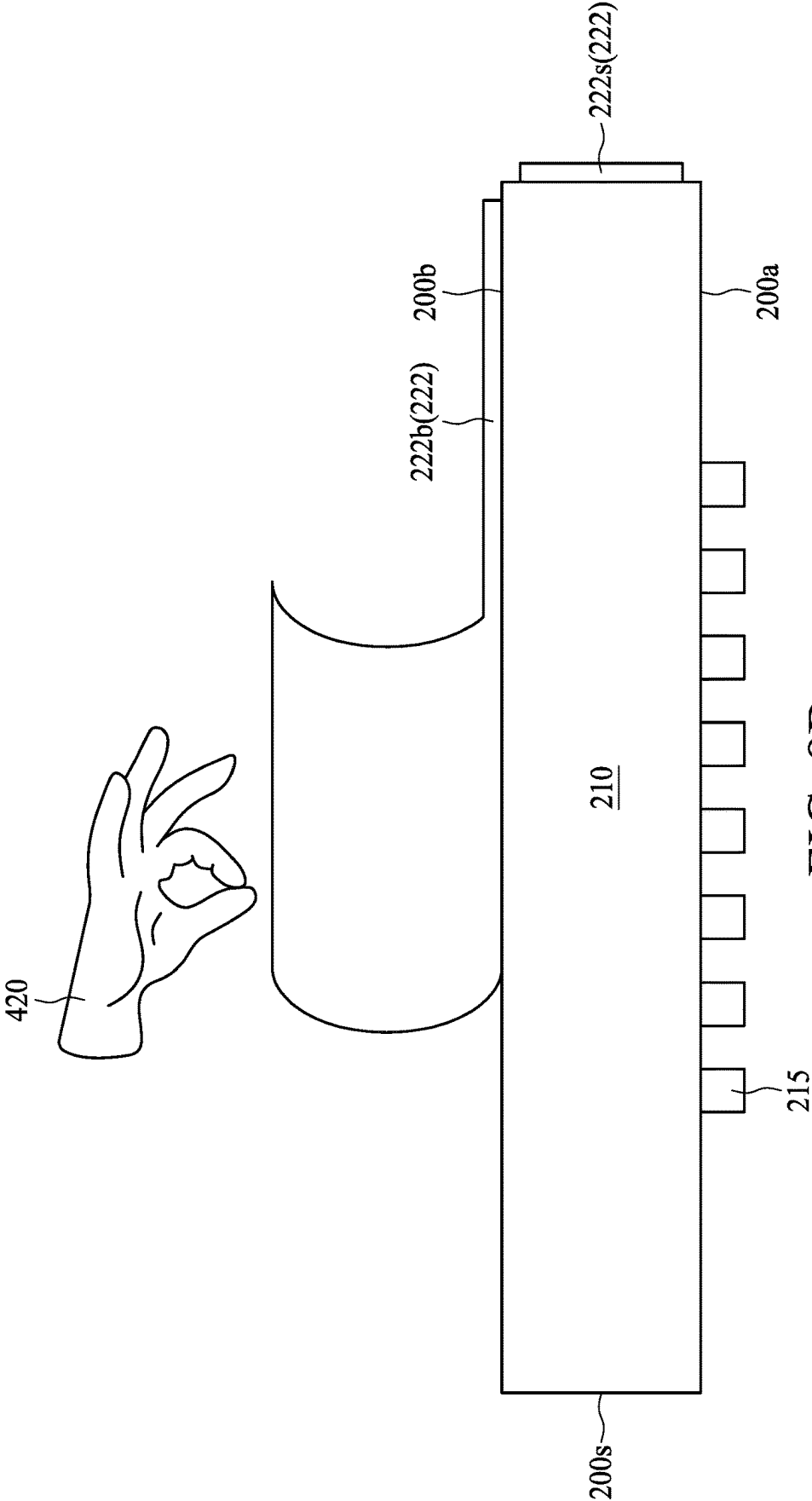


FIG. 8B

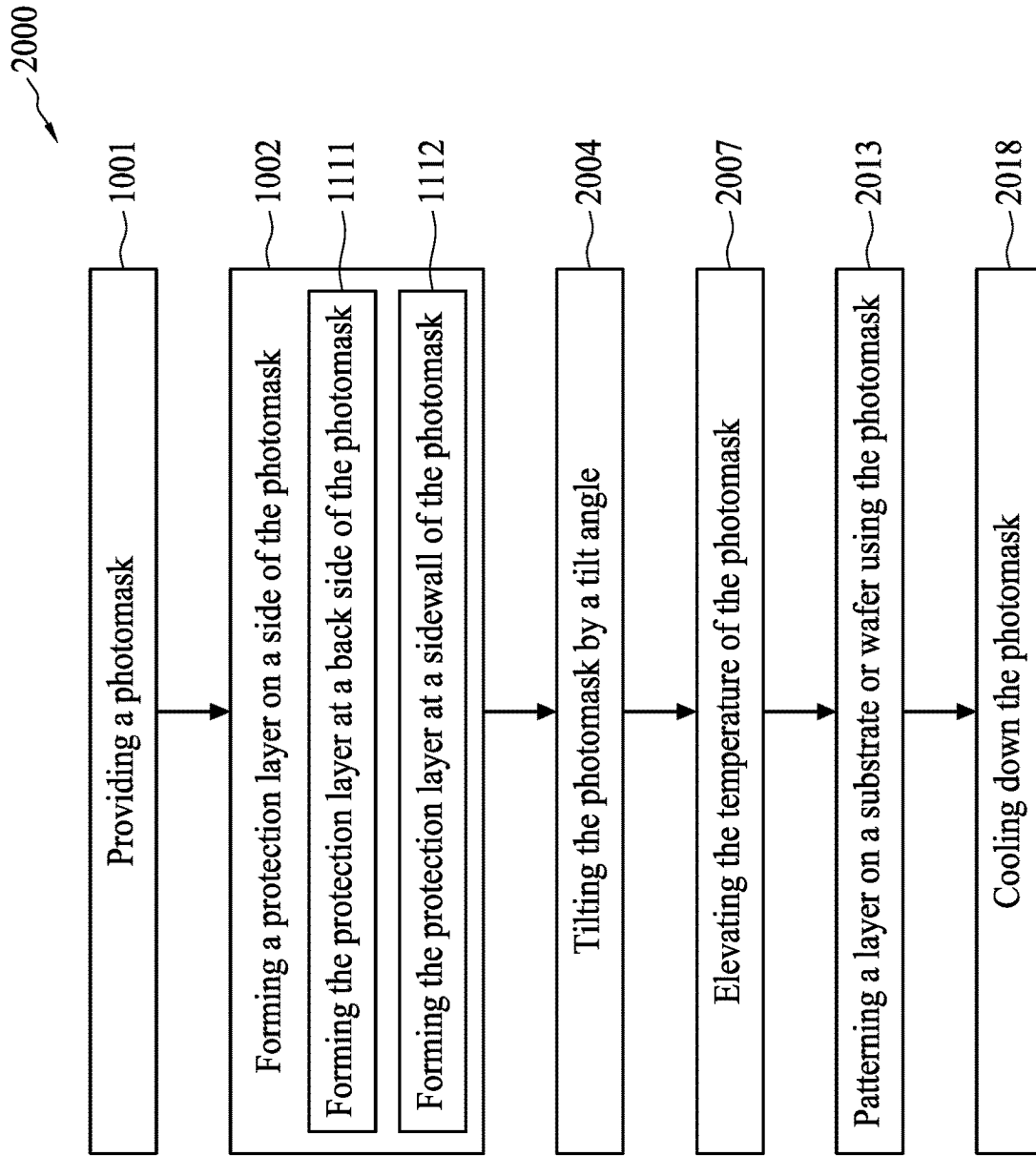


FIG. 9

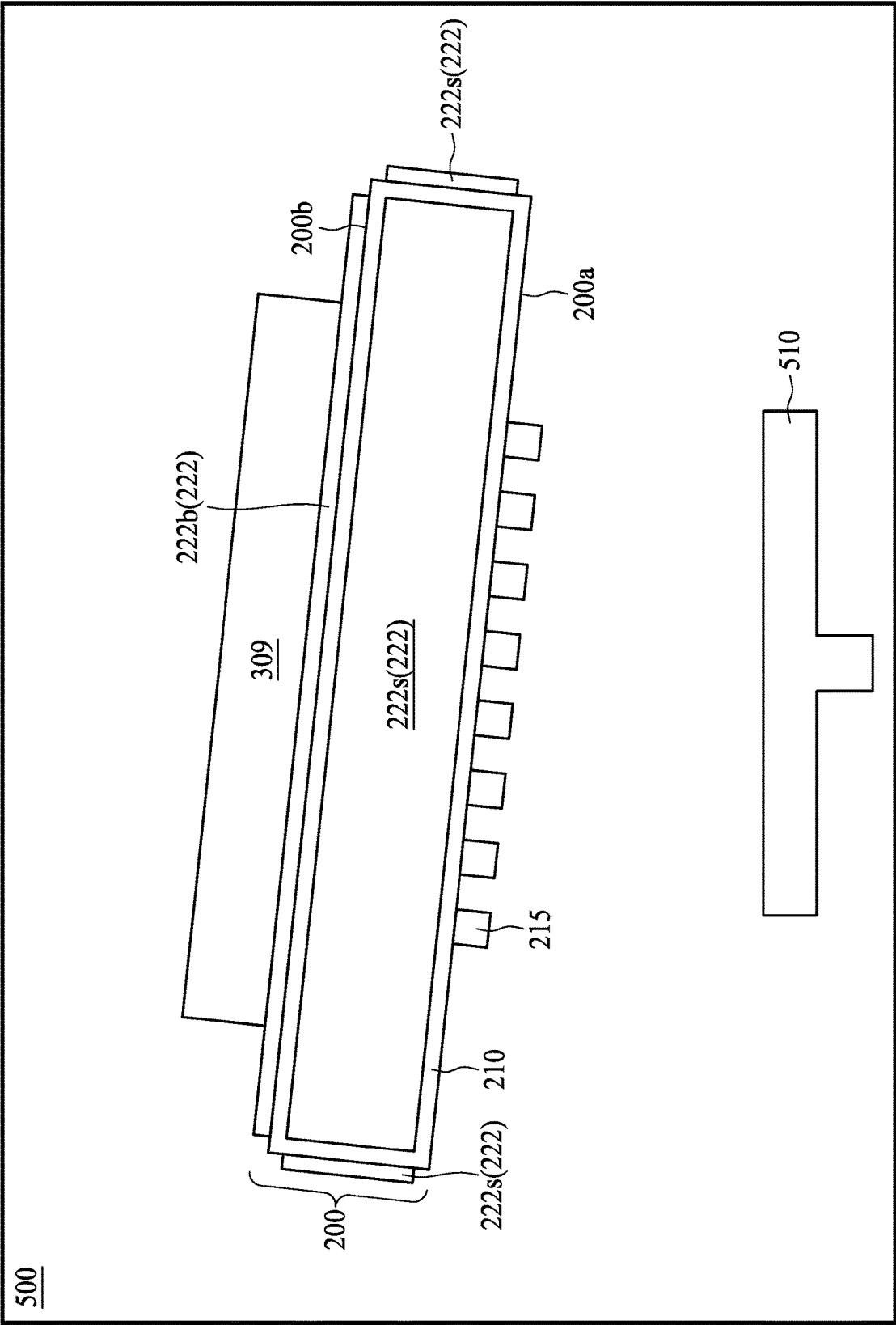


FIG. 10

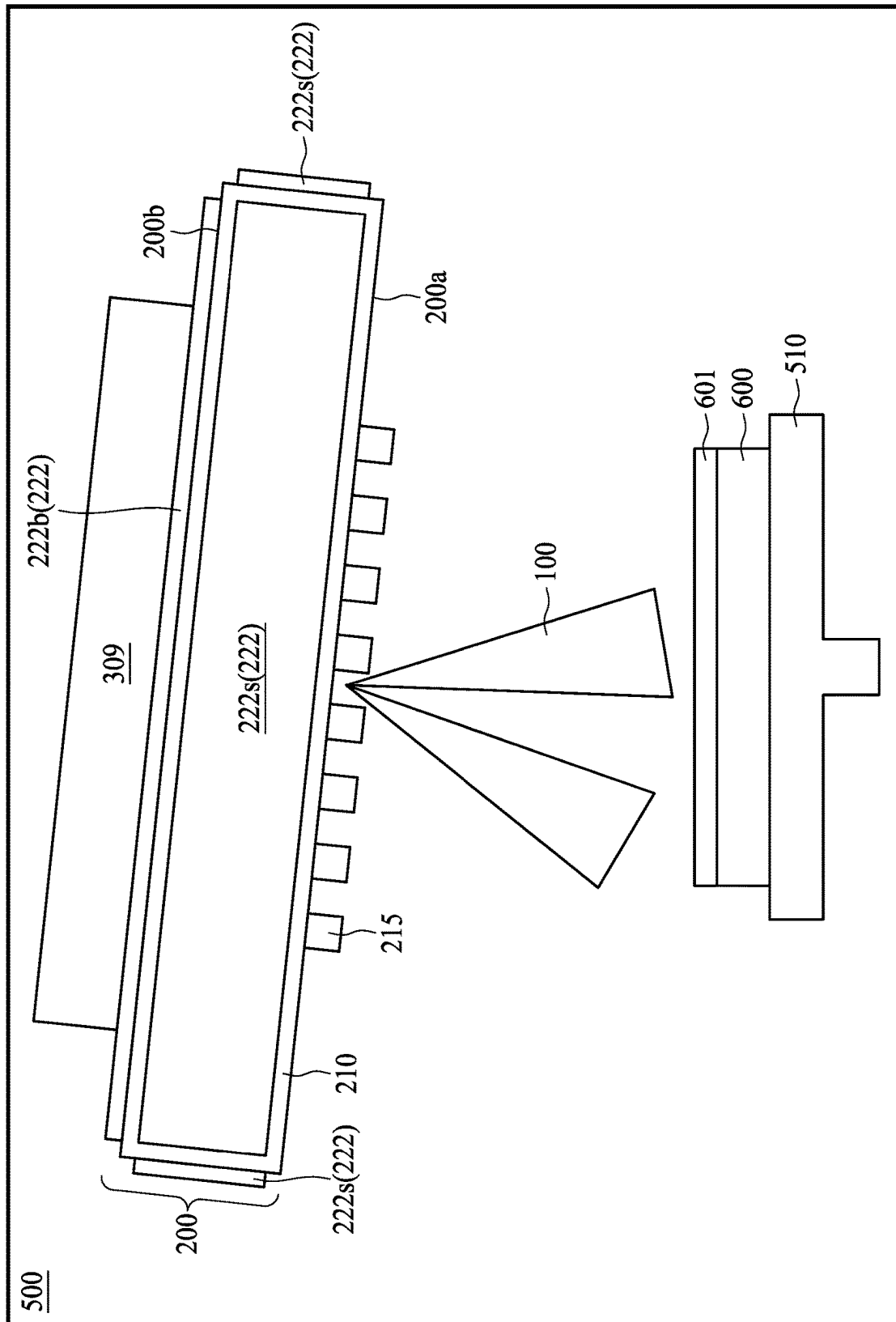


FIG. 11

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PHOTOMASK, METHOD OF FABRICATING A PHOTOMASK, AND METHOD OF FABRICATING A SEMICONDUCTOR STRUCTURE USING A PHOTOMASK

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of prior-filed U.S. non-provisional application Ser. No. 16/794,912, filed Feb. 19, 2020, which claims the benefit of prior-filed provisional application No. 62/827,488, filed Apr. 1, 2019, under 35 U.S.C. 120.

BACKGROUND

In the semiconductor industry, there is a trend toward higher device density. In order to achieve such higher density, smaller features are required. Such requirements likely involve scaling down device geometries to achieve lower fabrication costs, higher device integration density, higher speeds, and better performance. Along with the advantages from geometry size reductions, improvements to semiconductor devices are being made.

As semiconductor industry continues to evolve, advanced photolithography techniques have been widely used in integrated circuit fabrication operation. Photolithography operations may include techniques pertinent to coating a photoresist layer on a wafer and exposing the wafer to an exposing source.

Masks can be used in semiconductor fabrication operations to transfer a predetermined pattern onto a substrate. For example, after forming a layer over a substrate, the layer can be exposed to an actinic radiation through a mask, thus a pattern can be formed by subsequent development.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a front perspective view of a photomask and a chuck, according to some embodiments of the present disclosure.

FIG. 2 is a cross sectional view of a photomask and a chuck, according to some embodiments of the present disclosure.

FIG. 3 is a perspective view of a photomask, according to some embodiments of the present disclosure.

FIG. 4 is a schematic drawing showing a layered structure of a protection layer, according to some embodiments of the present disclosure.

FIG. 5 is a schematic drawing showing two states of a thermoplastic polymer layer of a protection layer, according to some embodiments of the present disclosure.

FIG. 6 shows a flow chart representing a method of fabricating a photomask, in accordance with some embodiments of the present disclosure.

FIG. 7 is a front perspective view of a photomask during intermediate stages of manufacturing operations, according to some embodiments of the present disclosure.

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FIG. 8A is a front perspective view of a photomask during intermediate stages of manufacturing operations, according to some embodiments of the present disclosure.

FIG. 8B is a front perspective view of a photomask during intermediate stages of manufacturing operations, according to some embodiments of the present disclosure.

FIG. 9 shows a flow chart representing a method of fabricating an integrated circuit, in accordance with some embodiments of the present disclosure.

FIG. 10 is a schematic drawing illustrating an apparatus during intermediate stages of lithography operation, according to some embodiments of the present disclosure.

FIG. 11 is a schematic drawing illustrating an apparatus during intermediate stages of lithography operation, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately,” or “about” generally means within a value or range which can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately,” or “about” means within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by

the terms “substantially,” “approximately,” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

During a photolithography operation, such as an extreme ultra-violet (EUV) photolithography operation, photomask may be carried by an electrostatic chuck with an adjustable angle facing radiation in order to complete the reflective mode of the photolithography operation. However, particles or scratches at a backside of the photomask (i.e., the side of the photomask contacting the electrostatic chuck) may weaken the control accuracy of adjustable angle of the photomask, and/or may weaken the carrying force between the photomask and the electrostatic chuck, which may cause the photomask detaching from the electrostatic chuck. Furthermore, particles may be attached to a surface of the photomask, and such particles may deteriorate the performance of lithography operations.

Conventionally, particles on a photomask may be removed by cleaning with wet chemical or by wiping. However, such procedures are time- and labor-consuming, and additional particles may fall on the photomask subsequent to cleaning/wiping operations. Another conventional operation is by placing a photomask in a sealing pod to prevent particles contamination. However, during the operations between taking out the photomask from the sealing pod and before lithography operations, particles may inevitably attached to the photomask causing contamination. Furthermore, the cleaning/wiping operations and sealing pod operations may not remedy scratches created on the photomask.

One lithography technique is extreme ultraviolet lithography (EUVL). The EUVL employs scanners using light in the extreme ultraviolet (EUV) region, having a wavelength of about 1-100 nm. EUV scanners use reflective rather than refractive optics, i.e., mirrors instead of lenses. The EUV lithography system includes a photomask stage configured to secure a photomask. In some embodiments, the photomask stage includes an electrostatic chuck (e-chuck) to secure the photomask. This is because that gas molecules absorb EUV light and the lithography system for the EUV lithography patterning is maintained in a vacuum environment to avoid the EUV intensity loss. By using reflective optics in the EUV system, the e-chuck holding the photomask is often controlled to tilt in a predetermined angle with respect to the underlying substrate to be exposed. Particles attached to the EUV photomask not only reduce the electrostatic force between the photomask and the e-chuck but also alter the preset tilting angle, rendering unsatisfactory exposure result. Therefore, particles on the EUV photomask are a sensitive factor affecting the performance of the EUV lithography system.

The present disclosure provides a photomask and a method of fabricating a photomask. The present disclosure also provides a method of forming an integrated circuit by using a photomask.

Referring to FIG. 1 and FIG. 2, FIG. 1 is a front perspective view of a photomask and a chuck, FIG. 2 is a cross sectional view of a photomask and a chuck, according to some embodiments of the present disclosure. A photomask 200 at least includes a patterned layer 215, a substrate 210, and a protection layer 222. The photomask 200 may

further include a reflective layer 250. In some embodiments, the photomask 200 is a reflective photomask, such as an extreme ultraviolet (EUV) photomask, or the like.

The photomask 200 has a front side 200a and a backside 200b opposite to the front side 200a, wherein the front side 200a of the photomask 200 has the patterned layer 215. The photomask 200 further has a sidewall 200s connecting the front side 200a and the back side 200b. In some of the embodiments, the substrate 210 of the photomask 200 has a rectangular shape, and the photomask 200 has four sidewalls 200s, but the present disclosure is not limited thereto. The substrate 210 of the photomask 200 may include a low thermal expansion material (LTEM) or a silicon-based material, such as quartz. The reflective layer 250 is over the substrate 210 and between the front side 200a and the back side 200b of the photomask 200. The reflective layer 250 includes reflective material to reflect a radiation irradiated on the front side 200a of the photomask 200. In some embodiments, the reflective layer 250 includes metal. In some embodiments, the reflective layer 250 includes multi-material layer, such as multi-metal layer, alternating molybdenum and silicon (Mo—Si) layers, or the like. For example, a reflective layer 250 of an EUV photomask may include about 40 pairs to about 50 pairs of molybdenum and silicon, and the reflective layer 250 may have an EUV light reflectivity of up to 67% at 13.5 nm wavelength.

In some embodiments, the photomask 200 can further be secured by a chuck 309 during a photolithography operation. The chuck 309 can be an electrostatic chuck (E-chuck) which utilizes electrostatic force to secure the photomask 200, for example, coulomb force type chuck, Johnson-Rahbeck force type chuck (which can be referred to as J-R chuck), gradient force type chuck, positively charged chuck, negatively charged chuck, or the like. In some of the embodiments, the photomask 200 may be secured by the chuck 309 in a tilted manner, wherein the tilt angle may be adjustable by the chuck 309.

Specifically, when some particles falls on the back side 200b of the photomask 200, or when scratches is created on the back side 200b of the photomask 200, an electrostatic force between the photomask 200 and the chuck 309 may be weakened due to reduced contact area between the photomask 200 and the chuck 309. In addition, a tilt angle of the photomask 200 may be affected by particles or scratches on the back side 200b of the photomask 200. Furthermore, the particles on a surface of the photomask 200 may cause contamination, and scratches may hinder the removal of particles.

In order to facilitate the removal of particles and/or improve the recovery of the scratches, one or more protection layer 222 is disposed on a side of the photomask 200. Referring to FIG. 1, FIG. 2 and FIG. 3, FIG. 3 is a perspective view of a photomask, according to some embodiments of the present disclosure. At least one protection layer 222 is disposed on a side of the photomask 200. Specifically, a protection layer 222 is disposed on the back side 200b of the photomask 200 (which can also be referred to as a substrate side) and/or disposed on the sidewall 200s of the photomask 200. For example, as shown in FIG. 3, a protection layer 222b is disposed on the back side 200b and a plurality of protection layers 222s are respectively disposed on each of the sidewall 200s. In the case of the photomask 200 having four sidewalls 200s connecting between the front side 200a and the back side 200b, four protection layers 222s are respectively on four sidewalls 200s of the photomask 200. However, the present disclosure is not limited thereto, wherein the side of the photomask 200

at least partially covered by a protection layer **222** is optional. The material of the protection layer **222b** on the back side **200b** of the photomask **200** and the protection layer **222s** on the sidewall **200s** of the photomask **200** may, or may not be identical. The material of the protection layer **222** (which may include the protection layer **222b** and/or the protection layer **222s**) will be subsequently discussed in FIG. 4 and FIG. 5. Note that hereinafter the protection layer **222h** and the protection layer **222s** may be collectively deemed as the protection layer **222**.

Since the particles on the protection layer **222** may travel from a back side **200b** to a sidewall **200s** of the photomask **200** as a result of the electric field generated by the e-chuck, the configuration of a gap G (i.e. the discontinuity of the protection layer **222**) may hinder the particles from traveling across different sides of the photomask **200** by prohibiting the particles from crossing the gap G, thereby reducing the risk of particles falling on the patterned layer **215** or contaminating substrates or wafers during lithography operations. Thus, in some embodiments, if the protection layer **222** is disposed on more than one side of the photomask **200**, each of adjacent protection layers **222** on adjacent sides are discontinuous and physically separated. In some embodiments, the gap G is at least 1 mm in order to effectively prohibiting the particles from crossing the gap G. If the gap G is less than a certain distance (such as mm), the gap G may not effectively hinder the particles from traveling across different sides of the photomask **200**. Alternatively stated, a portion of the back side **200b** and/or a portion of the sidewall **200s** is exposed from the protection layer **222**. For example, a gap G is between the protection layer **222b** on the back side **200b** and a protection layer **222s** on the sidewall **200s**. For another example, a gap C is between a protection layer **222s** on the sidewall **200s** and another protection layer **222s** on the sidewall **200s**.

Furthermore, since the protection layer **222** may be expanded under an elevated temperature when performing a lithography operation, the gap G may provide an adequate space for expansion in order to prevent one protection layer **222** to contact an adjacent protection layer **222**. A width of the gap G is wide enough so as to prevent merging of the adjacent protection layers **222** under the elevated temperature. For example, the photomask **200** may be elevated to in a range from about 50 to about 60 degree Celsius during an EUV lithography operation, and the gap G between two protection layer **222** may not close up when the photomask **200** is below 80 degree Celsius. In some embodiments, in the case of a length of a side of the photomask **200** being around 15.2 cm, a length of a side of the protection layer **222** may be around 15 cm.

Referring to FIG. 4, FIG. 4 is a schematic drawing showing a layered structure of a protection layer, according to some embodiments of the present disclosure. The protection layer **222** may include polymer. In some embodiments, the protection layer **222** may have a multi-layer structure. In some embodiments, the protection layer **222** may include an adhesive layer **222w** (which may be the side attached to a surface of the photomask **200**), a thermoplastic layer **222y** over the adhesive layer **222w**, and a hydrophobic layer **222z** over the thermoplastic layer **222y**. In some embodiments, the protection layer **222** may optionally further include a body layer **222x** between the adhesive layer **222w** and the thermoplastic layer **222y**.

In some embodiments, the adhesive layer **222w** may include polymer with proper cohesive strength, such as acrylic, styrene ethylene styrene (SEBS), polyethylene (PE), pressure-sensitive materials, the derivative thereof, the com-

bination thereof, or the like. It should be noted that the adhesive layer **222w** provide a sufficient cohesive strength so that the protection layer **222** does not unintentionally detached from the photomask **200** at least during a lithography operation. The adhesive layer **222w** may further be intentionally detached from the photomask **200** with a proper removal operation, either by manual removal or by robotic removal, such as pulling the protection layer **222** from an edge. Furthermore, the aforesaid material of the adhesive layer **222w** may leave less glue residue on the photomask **200** (which may be made by quartz) comparing to other types of materials.

Since the photomask **200** may be secured by an electrostatic chuck, electrostatic discharge (ESD) may be induced. Specifically, electrical overstress transient may produce high peak power and/or high electrical field that can destroy circuits having sensitive component. The electrical overstress transient may result from electromagnetic pulse, electrostatic discharge, or lightening, wherein such transient may be increased to a peak magnitude in a short period of time. In order to alleviate the issue stems from electrostatic discharge, the adhesive layer **222w** may further include conductive materials. For example, the adhesive layer **222w** may be doped with conductive dopant. Specifically, such conductive materials are configured to clamp the transient voltage to a lower value in the duration of electrical overstress transient induction. In response to a greater electrical overstress transient, the adhesive layer **222w** may rapidly convert to a lower resistance state; and when the electrical overstress transient is alleviated, the adhesive layer **222w** may convert back to a higher resistance state. For example, by utilizing conductive dopant, such as metallic ion (e.g. silver ion), graphene, or the like, may raise the electrical conductivity of the adhesive layer **222w**, and may clamp the transient voltage to a lower value. Thereby the risk of inducing electrostatic discharge may be mitigated. Furthermore, the adhesive layer **222w** having conductive material may alleviate static electricity accumulation, thereby contribute to ESD prevention.

In some embodiments, the body layer **222x** may be optionally included in the protection layer **222**, between the adhesive layer **222w** and the thermoplastic layer **222y**, in order to improve mechanical strength and the reliability of the protection layer **222**. In some embodiments, the body layer **222x** may improve extensibility of the protection layer **222**.

Referring to FIG. 4 and FIG. 5, FIG. 5 is a schematic drawing showing two states of a thermoplastic polymer layer of a protection layer, according to some embodiments of the present disclosure. In some embodiments, the thermoplastic layer **222y** includes polymer, such as thermoplastic polyurethanes (TPU), polyvinylidene fluoride (PVDF), the derivatives thereof, the combination thereof, or other suitable materials used as self-healing materials, stimuli responsive material, or shape memory materials. In some embodiments, the polymer of the thermoplastic layer **222y** may include cross-linked structure which determines one state of shape, and/or include molecular switches having a state cycle, such as a cycle of crystalline state and non-crystalline state (or amorphous state). In some embodiments, the polymer of the thermoplastic layer **222y** may include blocks polymer responsive to temperature changes.

Specifically, as shown in FIG. 5, the polymer of the thermoplastic layer **222y** is under state A under a first temperature (such as room temperature or ambient temperature). Under state A, an arrangement of the polymer can be deemed as in crystalline state, which may have crystalline

polymer blocks **2220**. For example, the polymer may include amide group ($-\text{CONH}$), carbonyl group ($-\text{CO}$), or difluoromethylene group ($-\text{CF}_2$). If the protection layer **222** is macroscopically being pressured, squeezed, or scratched microscopically, the number of crystalline polymer blocks **2220** will be increased from a first equilibrium state to a second equilibrium state. If a temperature of the thermoplastic layer **222y** is elevated (which may be heated to in a range from about 50 degree Celsius to about 60 degree Celsius by the photomask **200** during lithography operation, as will be subsequently discussed in FIG. **11**), the polymer of the thermoplastic layer **222y** is converted to state B, and the polymer blocks **2220** of the thermoplastic layer **222y** is relaxed out and became multiple relaxed polymers **2221**. Under state B, an arrangement of the polymer can be deemed as in non-crystalline state, or in amorphous state. Alternatively stated, a conversion temperature of the polymer of the thermoplastic layer **222y** between state A (crystalline state) and state **13** (non-crystalline state) can be in a range from about 50 degree Celsius to about 60 degree Celsius, or, lower than an operation temperature of a given photomask under a given lithography operation but greater than room temperature or ambient temperature, thereby the polymer of the thermoplastic layer **222y** can convert from crystalline state to non-crystalline state during the lithography operation. In some embodiments, when the thermoplastic layer **222y** is not heated to be above a certain temperature (such as 50 degree Celsius), the polymer blocks **2220** of the thermoplastic layer **222y** may not be effectively relaxed.

If the thermoplastic layer **222y** is cooled down from the range of about 50 degree Celsius to about 60 degree Celsius (or an operation temperature of a given photomask) to around room temperature (or ambient temperature), the polymer of the thermoplastic layer **222y** is converted from State B (non-crystalline state) to State A (crystalline state). The relaxed polymers **2221** are rearranged to crystalline polymer blocks **2220**, thereby the thermoplastic layer **222y** may substantially recover to the original state with crystalline polymer blocks **2220** under the aforesaid first equilibrium state, and the scratches created on the protection layer **222** may be substantially, or at least partially, removed.

It should be noted that herein in the present disclosure, the difference of state A (crystalline state) and state B (non-crystalline state) can be defined by respective crystallinities, depending on which material is utilized in the thermoplastic layer **222y**.

In some embodiments, polymer that includes amide group, carbonyl group, difluoromethylene group, or the like, may possess similar characteristic of conversion between different crystallinity states by temperature change. As previously discussed, thermoplastic polyurethanes (TPU) and polyvinylidene fluoride (PVDF) are two of the examples. In some embodiments, additive such as plasticizers can be optionally added in the thermoplastic layer **222y**.

Referring back to FIG. **4**, the hydrophobic layer **222z** is over the thermoplastic layer **222y**. The hydrophobic layer **222z** is a coating layer that has hydrophobic tendency on its exposed surface, and/or has a lower adhesive force with regard to foreign particles. For example, the hydrophobic layer **222z** may include polymer, such as polyvinylidene fluoride (PVDF), polyurethane (PU), polypropylene (PP), polyvinyl chloride (PVC), the derivatives thereof, the combination thereof, or the like. Conventional quartz plate includes hydroxyl groups, which have a greater tendency to attract particles on its surface than the hydrophobic layer **222z** described herein. In order to facilitate the removal of particles over the photomask **200**, incorporating the hydro-

phobic layer **222z** in the protection layer **222** may reduce the attraction force between the particle and the contact surface, while providing a hydrophobic surface that facilitates the deionized water (DI water) cleaning of the particles on the protection layer **222**.

Referring to FIG. **6**, FIG. **6** shows a flow chart representing a method of fabricating a photomask, in accordance with some embodiments of the present disclosure. The method **1000** for fabricating a photomask includes providing a photomask (operation **1001**, which can be referred to FIG. **7**), and forming a protection layer on a side of the photomask (operation **1002**, which can be referred to FIG. **8A** and FIG. **8B**). Operation **1002** further includes at least one of the sub-operation forming the protection layer at a back side of the photomask (sub-operation **1111**), and forming the protection layer at a sidewall of the photomask (sub-operation **1112**).

Referring to FIG. **7**, FIG. **7** is a front perspective view of a photomask during intermediate stages of manufacturing operations, according to some embodiments of the present disclosure. A photomask **200** is provided, wherein the photomask **200** at least includes a patterned layer **215**, a substrate **210**, and a protection layer **222**. In some embodiments, the photomask **200** is a reflective photomask, such as an extreme ultraviolet (EUV) photomask, or the like. The details of the photomask **200** can be referred to the previous discussion with regard to FIG. **1**.

Referring to FIG. **7**, FIG. **8A** and FIG. **8B**, FIG. **8A** is a front perspective view of a photomask during intermediate stages of manufacturing operations, FIG. **8B** is a front perspective view of a photomask during intermediate stages of manufacturing operations, according to some embodiments of the present disclosure. Subsequently, at least one protection layer **222** is formed on a side of the substrate **210** of the photomask **200**, wherein an adhesive side of the protection layer **222** (e.g. adhesive layer **222w**) is attached to a side of the photomask **200**. The protection layer **222** can be selectively formed on the back side **200b** and/or one or more sidewall **200s** of the photomask **200**. The material of the protection layer **222** on the back side **200b** of the photomask **200** (which is denoted as protection layer **222b**) and the protection layer **222s** on the sidewall **200s** of the photomask **200** (which is denoted as protection layer **222s**) may, or may not be identical.

As shown in FIG. **8A**, in some embodiments, forming the protection layer **222** includes robotic application, for example, which may utilize a robotic arm **410** to attach the protection layer **222** on one or more side of the photomask **200**. The robotic application may be semi-automatically or automatically performed. As shown in FIG. **8B**, in some alternative embodiments, forming the protection layer **222** includes manual application, for example, which may require a user **420** to attach the protection layer **222** on one or more side of the photomask **200**. In some embodiments, the protection layer **222** may optionally be pressed against the photomask **200**, thereby the adhesion force between the photomask **200** and the protection layer **222** is increased.

Referring to FIG. **3**, FIG. **8A** and FIG. **8B**, Since the particles on the protection layer **222** may travel from a back side **200b** to a sidewall **200s** of the photomask **200** as a result of the electric field generated by the e-chuck, the configuration of a gap G (i.e. the discontinuity of the protection layer **222**) may hinder the particles from traveling across different sides of the photomask **200** by prohibiting the particles from crossing the gap G, thereby reducing the risk of particles falling on the patterned layer **215** or contaminating substrates or wafers during lithography operations.

In some embodiments, the protection layer **222** is fabricated prior to forming the protection layer **222** on a side of the photomask **200**. The operations of fabricating protection layer **222** includes forming an adhesive layer **222w**, forming a thermoplastic layer **222y** over the adhesive layer **222w**, and forming a hydrophobic layer **222z** over the thermoplastic layer **222y**. The operations of fabricating protection layer **222** may optionally include forming a body layer **222x** between the thermoplastic layer **222y** and the adhesive layer **222w**. Subsequently, the entire protection layer **222** including the adhesive layer **222w**, thermoplastic layer **222y**, the hydrophobic layer **222z**, and/or the body layer **222x** is attached to the photomask **200**, wherein the adhesive layer **222w** contacts with the photomask **200**. In some embodiments, the adhesive layer **222w** may include polymer with proper cohesive strength, such as acrylic, styrene ethylene styrene (SEBS), polyethylene (PE), pressure-sensitive materials, the derivative thereof, the combination thereof, or the like. The adhesive layer **222w** may further be doped with conductive materials, such as metallic ion (e.g. silver ion), graphene, or the like. In some embodiments, the thermoplastic layer **222y** includes polymer, such as thermoplastic polyurethanes (TPU), polyvinylidene fluoride (PVDF), the derivatives thereof, the combination thereof, or other suitable materials used as self-healing materials, stimuli responsive material, or shape memory materials. In some embodiments, the hydrophobic layer **222z** may include polymer, such as polyvinylidene fluoride (PVDF), polyurethane (PU), polypropylene (PP), polyvinyl chloride (PVC), the derivatives thereof, the combination thereof, or the like. In some embodiments, the body layer **222x** includes polymer.

In alternative embodiments, the adhesive layer **222w**, the body layer **222x**, thermoplastic layer **222y**, the hydrophobic layer **222z** (or any combination thereof) may be sequentially applied on a side of the photomask **200**.

Referring to FIG. 9, FIG. 9 shows a flow chart representing a method of fabricating an integrated circuit, in accordance with some embodiments of the present disclosure. The method **2000** for fabricating an integrated circuit includes providing a photomask (operation **1001**, which can be referred to FIG. 7), forming a protection layer on a side of the photomask (operation **1002**, which can be referred to FIG. 8A and FIG. 8B), tilting the photomask by a tilt angle (operation **1004**, which can be referred to FIG. 10), elevating the temperature of the photomask (operation **1007**, which can be referred to FIG. 11), patterning a layer on a substrate or wafer using the photomask (operation **1013**, which can be referred to FIG. 11), and cooling down the photomask (operation **1018**, which can be referred to FIG. 11). Operation **1002** further includes at least one of the sub-operation forming the protection layer at a back side of the photomask (sub-operation **1111**), and forming the protection layer at a sidewall of the photomask (sub-operation **1112**), as previously discussed in FIG. 6, FIG. 8A, and FIG. 8B.

Referring to FIG. 10, FIG. 10 is a schematic drawing illustrating an apparatus during intermediate stages of lithography operation, according to some embodiments of the present disclosure. Subsequent to forming a protection layer **222** on a side of the photomask **200** the photomask **200** is secured by a chuck **309** in an apparatus **500**. In some embodiments, the chuck **309** is an electrostatic chuck (E-chuck). Optionally, in some embodiments, a tilt angle of the photomask **200** is adjustable by the chuck **309**. The apparatus **500** may further include a supporter **510** to support a wafer or a substrate during a photolithography operation.

Referring to FIG. 11, FIG. 11 is a schematic drawing illustrating an apparatus during intermediate stages of lithography operation, according to some embodiments of the present disclosure. Subsequently, a layer **601** (e.g. a photoresist layer, a sacrificial layer, a patternable layer, or the like) is formed on a wafer **600**, and the wafer **600** is secured by the supporter **510**. Thence the layer **601** is patterned by the photomask **200** in a photolithography operation in the apparatus **500**. The photolithography operation includes heating the photomask **200**, which may include irradiating an actinic radiation **100** on the front side **200a** of the photomask **200**. In some embodiments, the actinic radiation **100** is extreme ultraviolet (EUV), and the actinic radiation **100** is reflected by the reflective layer **250** (as shown in FIG. 2) of the photomask **200**. Since the reflective layer **250** may include metal, a temperature of the photomask **200** as well as the protection layer **222** may be elevated. In some embodiments, the photomask **200** is heated to in a range from about 50 degree Celsius to about 60 degree Celsius. If the photomask **200** is overheated, for example, heated to above 60 degree Celsius, the patterned layer **215** may be unacceptably missized or distorted, thus a temperature of the photomask **200** is controlled to be under 60 degree Celsius, or under a predetermined temperature based on the requirement of the lithography operation or the type of the actinic radiation **100**. The layer **601** is thereby patterned. In some embodiments, development operations are subsequently performed in order to fabricate integrated circuit.

It should be noted that, since the protection layer **222** may be expanded under an elevated temperature when performing a lithography operation, the gap **G** (as shown in FIG. 3) between two protection layers **222** created as a result of forming protection layers **222** on the photomask **200** (as discussed in FIG. 8A and FIG. 8B) may provide an adequate space for expansion and thereby prevent one protection layer **222** to contact an adjacent protection layer **222**. A width **G** of the gap is wide enough so the gap **G** may not merged under the operable temperature of the photomask **200** under lithography operation. For example, the photomask **200** may be elevated to in a range from about 50 to about 60 degree Celsius during the EUV lithography operation, and the gap **G** between two protection layers **222** may not close up when the photomask **200** is below 80 degree Celsius. In some embodiments, the apparatus **500** may optionally further include a temperature monitor (not shown in FIG. 11) to detect a temperature of the photomask **200**.

As previously discussed in FIG. 4 and FIG. 5, the polymer of the thermoplastic layer **222y** converts from state A (crystalline state) to state B (non-crystalline state), wherein a conversion temperature of the polymer of the thermoplastic layer **222y** between state A (crystalline state) and state B (non-crystalline state) can be in a range from about 50 degree Celsius to about 60 degree Celsius, or, lower than an operation temperature of a given photomask under a given lithography operation but greater than room temperature or ambient temperature. The polymer blocks **2220** (as well as non-balanced polymer blocks caused by scratch) of the thermoplastic layer **222y** is relaxed out and became multiple relaxed polymers **2221** with smaller sizes.

After a certain cycle of lithography operation, the irradiation of the actinic radiation **100** is halted, and the photomask **200** as well as the protection layer **222** is cooled down from the operation temperature (in a range from about 50 degree Celsius to about 60 degree Celsius) to around room temperature (or ambient temperature). Thereby the

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polymer of the thermoplastic layer 222_y is converted from State B (non-crystalline state) to State A (crystalline state). The relaxed polymers 222₁ are rearranged to crystalline polymer blocks 222₀, and the thermoplastic layer 222_y may substantially recover to the original state with crystalline polymer blocks 222₀ under the aforesaid first equilibrium state, and the scratches created on the protection layer 222 may be substantially, or at least partially, removed.

Optionally, in some embodiments, after removing the photomask 200 from the apparatus 500, the protection layer 222 of the apparatus 500 can be cleaned. For example, the particles on the protection layer 222 can be cleaned by applying deionized water thereon.

Optionally, in some embodiments, the protection layer 222 can be replaced by another protection layer 222. The removal of the to-be-replaced protection layer 222 from the photomask 200 can either be removed by manual or by robotic. The removal of the protection layer 222 may include pulling the protection layer 222 from an edge. The attachment of another protection layer 222 to the photomask 200 can either be by manual application or by robotic application, as previously discussed in FIG. 8A or FIG. 8B. The robotic application and removal may be semi-automatically or automatically performed.

The present disclosure discloses a photomask 200, a protection layer 222 for a photomask 222, a method for forming a photomask 200, and a method for forming an integrated circuit. The adhesive layer 222_w may allow the protection layer 222 to be attached to the photomask 200 with proper adhesive strength, and alleviate issue of glue residue remaining on the photomask 200 when removing the protection layer 222. Furthermore, since the adhesive layer 222_w may be doped by conductive materials, the issue of electrostatic discharge caused by the chuck 309 may be alleviated. The hydrophobic layer 222_z of the protection layer 222 facilitate the removal of particles thereon, and the thermoplastic layer 222_y of the protection layer 222 recovers scratches by switching between crystalline state and non-crystalline state stimulated by changing temperature. By incorporating the protection layer 222 having the thermoplastic layer 222_y to a photomask 200 that a temperature thereof may be elevated during a photolithography operation, the scratches may be at least partially recovered during such photolithography operation, thus the reliability of the photomask 200 and the efficiency of recovering scratches are improved.

In addition, by having a gap G between adjacent protection layers 222, the moving particles may be stuck in the gap C or at an edge of the protection layer 222, and the contamination on wafers 600 caused by such particles may be alleviated. And the gap G is configured to provide a space for the adjacent protection layer 222 to expand without contacting each other.

By removing particles falls on the photomask 200 or recovering scratches on the photomask 200, an electrostatic force between the photomask 200 and the chuck 309 may be improved, thus alleviate the risk of photomask 200 undesirably detached from the chuck 309. In addition, the accuracy of adjusting a tilt angle of the photomask 222 can thereby be improved.

Some embodiments of the present disclosure provide a photomask, including a front side having a patterned layer, a back side opposite to the front side, a sidewall connecting the front side and the back side, a reflective layer between the front side and the back side, and a polymer layer on the backside of the photomask.

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Some embodiments of the present disclosure provide a protection layer for extreme ultra-violet (EUV) photomask, including an adhesive layer, a thermoplastic layer over the adhesive layer, a hydrophobic layer over the thermoplastic layer, wherein the thermoplastic layer comprises polymer having a crystalline-non crystalline conversion temperature in a range of from about 50 to 60 degrees Celsius.

Some embodiments of the present disclosure provide a method for fabricating a photomask, including providing a photomask, comprising a substrate, a reflective layer, and a patterned layer, forming a protection layer at a substrate side of the photomask, wherein the protection layer comprising an adhesive layer attaching to a substrate side of the photomask.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other operations and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present invention. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

What is claimed is:

1. A photomask, comprising:

a substrate;
a patterned layer over a first surface of the substrate; and
a polymer multilayer over a second surface of the substrate, wherein the polymer multilayer comprises a thermoplastic polymer and a hydrophobic layer, wherein the thermoplastic polymer is between the hydrophobic layer and the second surface of the photomask,
wherein the polymer multilayer further comprises an adhesive layer between the thermoplastic polymer and the second surface of the substrate.

2. The photomask of claim 1, wherein the polymer multilayer is over a sidewall of the photomask.

3. The photomask of claim 1, wherein the polymer multilayer is over a backside of the photomask.

4. The photomask of claim 1, wherein the thermoplastic polymer comprises thermoplastic polyurethanes (TPU) or polyvinylidene fluoride (PVDF).

5. The photomask of claim 1, wherein the thermoplastic polymer comprises a crystalline-non crystalline conversion temperature in a range of from about 50 to 60 degrees Celsius.

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6. The photomask of claim 1, wherein the adhesive layer is between the thermoplastic polymer and a back side of the photomask.

7. The photomask of claim 1, wherein the adhesive layer comprises a conductive dopant.

8. The photomask of claim 7, wherein the conductive dopant consists a metallic ion or graphene.

9. The photomask of claim 1, wherein the adhesive layer is doped with silver ion.

10. A photomask, comprising:

a substrate;

a patterned layer over a first surface of the substrate; and
a first film over a second surface of the substrate, comprising:

an adhesive layer, wherein the adhesive layer comprises a conductive dopant consisting of a metallic ion or graphene;

a thermoplastic layer over the adhesive layer; and

a hydrophobic layer over the thermoplastic layer.

11. The photomask of claim 10, wherein the thermoplastic layer comprises polymer layer having a crystalline-non crystalline conversion temperature in a range of from about 50 to 60 degrees Celsius.

12. The photomask of claim 10, wherein the first film is over a first sidewall of the substrate.

13. The photomask of claim 12, further comprising a second film over a backside of the substrate.

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14. The photomask of claim 13, wherein a first corner of the substrate is exposed from the first film and the second film.

15. The photomask of claim 12, further comprising a third film over a second sidewall of the substrate, wherein the second sidewall is connected to the first sidewall.

16. The photomask of claim 15, wherein a second corner of the substrate is exposed from the first film and the third film.

17. A method for fabricating a semiconductor structure with an extreme ultraviolet (EUV) photomask, comprising:

placing a substrate over a supporter;

forming a photoresist layer over the substrate;

patterning the photoresist layer with an actinic radiation reflected off an EUV photomask, wherein a protective film is disposed on the EUV photomask, and the protective film comprises:

a thermoplastic polymer and a hydrophobic layer, wherein the thermoplastic polymer is between the hydrophobic layer and a surface of the photomask, wherein protective film further comprises an adhesive layer facing the photomask.

18. The method of claim 17, wherein the adhesive layer is doped with silver ion.

19. The method of claim 17, wherein the adhesive layer comprises a conductive dopant.

20. The method of claim 19, wherein the conductive dopant consists a metallic ion or graphene.

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